

ENEOPi

FRAGMENT SEGMENTATION

GIỚI THIỆU THÀNH VIÊN

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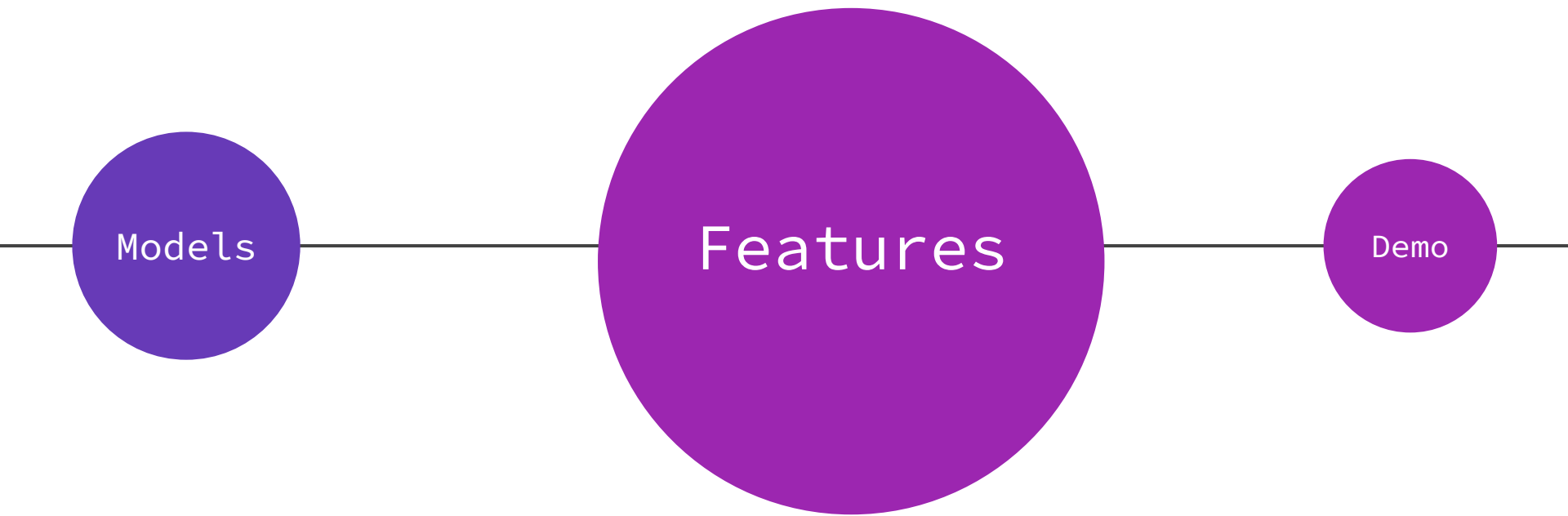
Phan Trọng Đài - 2312026

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Nguyễn Nam Anh - 23125076

Skills & expertise



Model

UNET - TOP 1 Kaggle

YOLOV8

YOLOV8 + DepthAnything

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UNET - Kaggle solution

Model: UNET++ with efficientnet-b5 encoder

Transfer learning: pretrained weight ImageNet

Augmentation: Basic transform with Albumentations

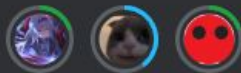
Loss: Binary Cross Entropy + Dice Loss

More: 200 epochs + AdamW + Cosine

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0.98355

16

2mo

Problem?

YOLOv8

Ultralytics API

Model	mAP50 Box	mAP50 Mask	Average Time (s)
v8n	0.769	0.755	0.3724
v8n - high tuning	0.845	0.835	0.4164
v8s	0.835	0.819	0.5323
v8m	0.855	0.840	0.7938

DepthAnythingV2 - YOLOv8 RGBD

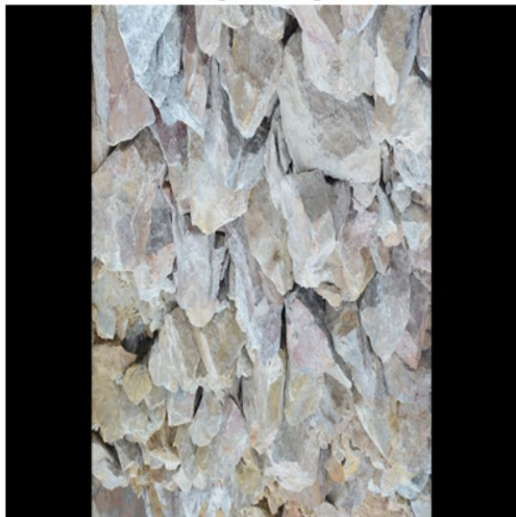
DepthAnythingV2: zero-shot

vits - ~2s

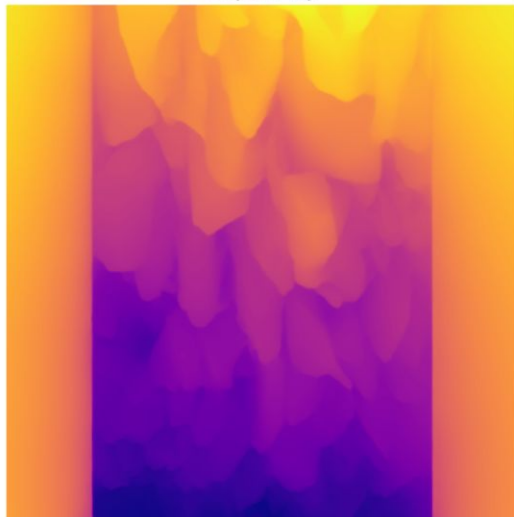
vitm - ~6s

vitl - ~12s

Original Image



Depth Map



DepthAnythingV2 - YOLOv8 RGBD

YOLOv8 only depth map -> Can't detect small fragments

YOLOv8 RGBD

- Input: Origin image + Depth map (from DepthAnythingV2)
- Ultralytics API

Model	mAP50 Box	mAP50 Mask	Average Time (s)
RGBD + v8n	0.886	0.879	0.4139
RGBD + v8s	0.916	0.908	0.5682
RGBD + v8m	0.941	0.931	0.7991

Feature

Two modes

Docker + ONNX

CDF

Sign in + oauth

Tech stack

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Two modes

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RGBD + v8n	0.886	0.879	~2s + 0.4139s
RGBD + v8s	0.916	0.908	~2s + 0.5682s
RGBD + v8m	0.941	0.931	~2s + 0.7991s

Two modes

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Mode: Fast ☐ Precise 1280.5ms



Mode: Fast ☒ Precise 2365.1ms



API design

Delegate work to client as much as possible:

- + The client downsample the image and convert to PNG before sending to the back-end
- + The client only receive a PNG compressed binary mask, and have to perform rendering + coloring by itself
- + Data transfer size: 32kb per fragment + PNG compression

ONNX + Docker

Ultralytics is the main API for YOLOv8

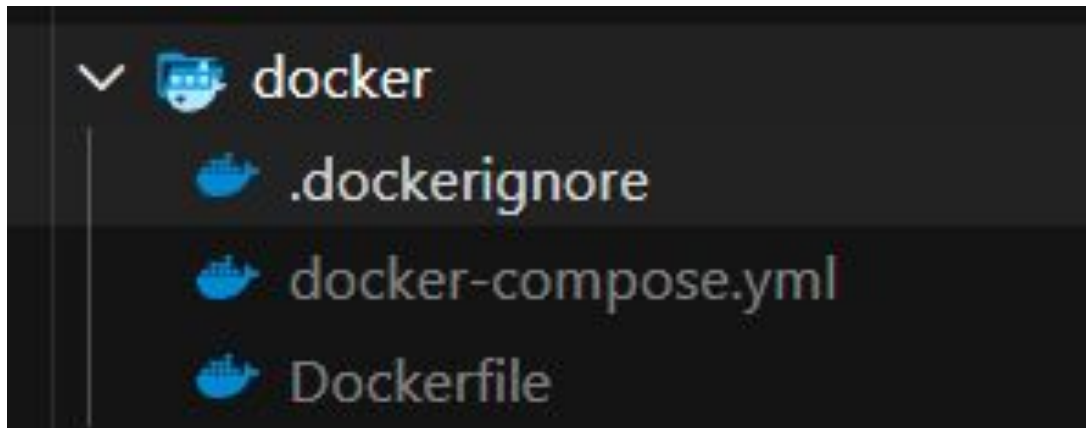
But Ultralytics is too heavy:

- + Hard for collaborate between teammates
- + Docker need ~20 mins to install only Ultralytics

=> Convert weight.pt to weight.onnx

ONNX + Docker

Support **DOCKER** for seamless deployment



ONNX + Docker

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But Ultralytics is too heavy:

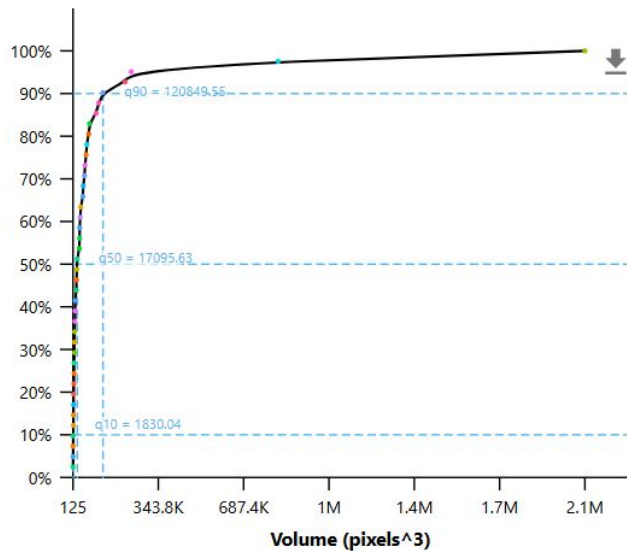
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CDF

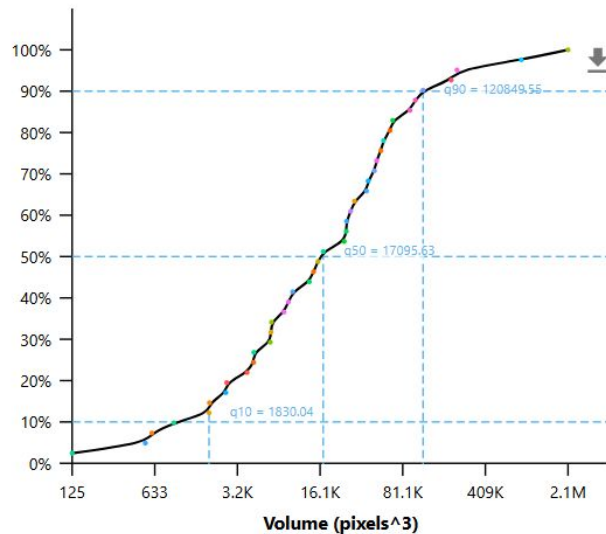
Scale: Linear ☐ Log

CDF



Scale: Linear ☒ Log

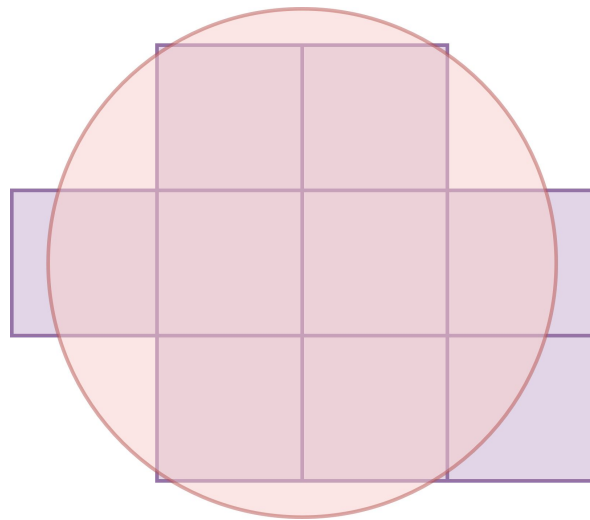
CDF



CDF - Volume estimation

Pixel based volume estimation:

- + Fast, simple
- + Model-agnostic
- + Can be implemented in the client
- Doesn't account for depth and perspective distortion



CDF - Volume estimation (with depth)

Idea: estimate the area of a visible, almost flat surface then extrude.

Procedure:

- Convert the depth mask into a point cloud
- Replace the smallest eigenvalue with the mean of the rest
- Compute the square root of the product
- This can be implemented with SVD and normalization

SIGN IN + OAUTH

Welcome Back

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Username or Email

Password

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Sign In

or

 Continue with Google

Don't have an account? [Sign up](#)

Join ENEOPI

Create your account to start using Blast Segmentation

Username

Email Address

Password

Confirm Password

Create Account

or

 Continue with Google

Already have an account? [Sign in](#)

Welcome, admin3!

Logout



Authentication Successful!

You have successfully signed in with Google.
Redirecting you to the application...



Tech Stack



Depth Anything V2



DEMO